

# Azure Monitor Private Link Scope for Isolated Telemetry

Tools, Azure services, usage, architecture, benefits, and key configuration points

## 1. Project Overview

Azure Monitor Private Link Scope (AMPLS) is used to provide private connectivity to Azure Monitor resources. The main goal is to send monitoring telemetry without relying on the public internet and to satisfy network-isolation requirements.

## 2. Azure Services and Tools Used

| Service / Tool                           | How it is used                                                              | How it helps                                        |
|------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------|
| Azure Monitor                            | Collects and monitors metrics, logs, alerts, and telemetry.                 | Provides overall monitoring and visibility.         |
| Azure Monitor Private Link Scope (AMPLS) | Defines the Azure Monitor resources available through private connectivity. | Creates a controlled private monitoring boundary.   |
| Azure Private Endpoint                   | Creates a private network interface in the VNet for the AMPLS connection.   | Provides a private IP-based connectivity.           |
| Virtual Network (VNet)                   | Hosts the application/VM and private endpoint.                              | Provides the private network boundary.              |
| Log Analytics Workspace                  | Stores and analyzes monitoring logs using KQL.                              | Supports log search, troubleshooting, and analysis. |
| Application Insights                     | Collects application requests, failures, performance, and other telemetry.  | Helps monitor application health.                   |
| Private DNS Zone                         | Resolves Azure Monitor private-link names to private IP addresses.          | Ensures traffic uses the private endpoint.          |
| Azure Monitor Agent (AMA)                | Collects telemetry from supported VMs and resources.                        | Provides a managed way to collect monitoring data.  |
| Data Collection Endpoint (DCE)           | Provides private endpoints for supported Azure Monitor agents.              | Helps data collection to telemetry ingestion.       |
| NSG / Firewall                           | Controls network traffic around the private subnet.                         | Adds another layer of network security.             |

## 3. How the Services Work Together

**Application / VM → Azure Monitor Agent → Virtual Network → Private Endpoint → AMPLS → Log Analytics / Application Insights → Azure Monitor**

The private endpoint receives a private IP address from the VNet. AMPLS connects the private endpoint to selected Azure Monitor resources. Private DNS helps clients resolve the relevant monitoring endpoints to the private IP instead of a public endpoint.

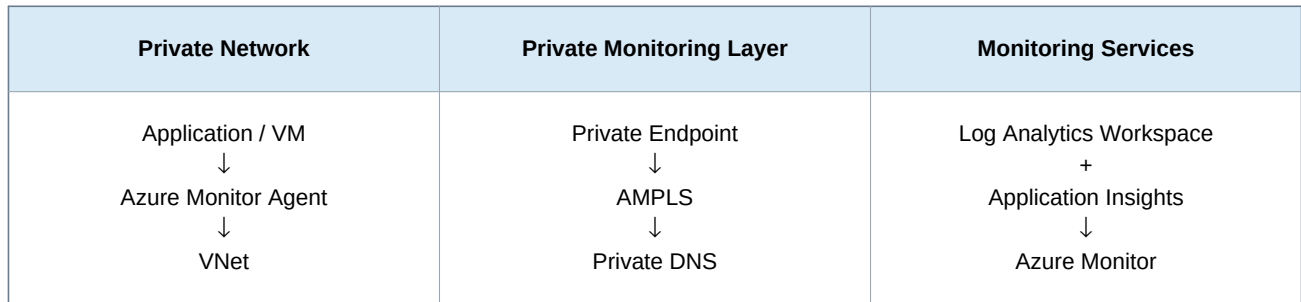
## 4. Key Components Explained

- **Azure Monitor:** Central monitoring platform for metrics, logs, alerts, and telemetry.
- **Log Analytics Workspace:** Stores monitoring logs and allows analysis with Kusto Query Language (KQL).
- **Application Insights:** Provides application-level telemetry such as requests, response times, failures, exceptions, and performance information.
- **AMPLS:** Defines which Azure Monitor resources can be reached through the private monitoring connection.
- **Private Endpoint:** Places a private network interface and private IP inside the VNet for the AMPLS connection.
- **Private DNS Zone:** Maps private-link DNS names to private IP addresses so applications can use the private endpoint.
- **Azure Monitor Agent:** Collects monitoring data from supported resources and sends it to Azure Monitor.

## 5. Why Private Link Scope Is Needed

- **Private telemetry:** Helps keep monitoring communication on private connectivity rather than depending on public internet access.
- **Network isolation:** Helps organizations meet requirements where monitoring resources must be accessed through a private network.
- **Controlled access:** AMPLS can define which Log Analytics workspaces and Application Insights components are reachable through the private connection.
- **Centralized monitoring boundary:** Provides a common private-link scope for the monitoring resources used by connected workloads.

## 6. Example Architecture



## 7. Configuration Flow

1. Create or select the Azure Monitor resources, such as a Log Analytics workspace and Application Insights component.
2. Create an Azure Monitor Private Link Scope (AMPLS).
3. Associate the required Azure Monitor resources with the AMPLS.
4. Create a Private Endpoint for the AMPLS in the required VNet/subnet.
5. Configure the required Private DNS zones and DNS links.
6. Configure monitoring agents and data collection according to the workload.
7. Apply appropriate NSG/firewall rules.
8. Test DNS resolution and telemetry flow, then verify that the required monitoring traffic follows the private configuration.

## 8. Important Configuration Considerations

- **AMPLS scope:** Carefully select the Log Analytics workspaces and Application Insights resources that need private access.
- **DNS:** Correct private DNS configuration is essential; a private endpoint alone does not automatically make every monitoring path private.
- **Public access settings:** Review public network access and access modes according to the required isolation level.
- **Multiple workspaces:** An AMPLS can be associated with multiple supported Azure Monitor resources, so scope design should be planned carefully.
- **Validation:** Do not assume that telemetry is private just because a private endpoint exists. Test DNS resolution, network connectivity, and the actual telemetry path.

## 9. Bottlenecks Mentioned in the Given Use Case

- 1. Private Link Scope configuration affects connected workspaces:** Scope and resource associations should be planned so that the correct monitoring resources are available to each network.
- 2. Partial configuration can leave public paths available:** DNS, AMPLS associations, public network access settings, and monitoring endpoints must be configured and validated together.

Reference: Microsoft Azure Monitor documentation on Private Link, AMPLS, private endpoints, and private DNS. Configuration details can vary by Azure Monitor resource and current Azure service capabilities.